

Chapter #5

Q no : 3

$$m = 1 \text{ kg}$$

$$a = 2 \text{ ms}^{-2}$$

$$\theta = 20^\circ$$

$$F_x = ? ; F_y = ?$$

$$c) \vec{F} = ?$$

Solution :

$$F_x = m a \cos \theta$$

$$= (1)(2) \cos 20$$

$$= 1.8793 \hat{i}$$

$$F_y = m a \sin \theta$$

$$= (1)(2) \sin 20$$

$$= 0.684 \hat{j}$$

$$\vec{F} = (1.87 \hat{i} + 0.68 \hat{j}) \text{ N}$$

Q no : 5

$$m = 120 \text{ kg}$$

$$F_1 = 32 \text{ N} \quad \theta_1 = 30^\circ$$

$$F_2 = 55 \text{ N} \quad \theta_2 = 0^\circ$$

$$F_3 = 41 \text{ N} \quad \theta_3 = -60^\circ$$

$$\vec{a}_{\text{ast}} = ? \quad ; \quad |\vec{a}| \quad ; \quad \theta_{\vec{a}_x} = ?$$

$$F_{1x} = 27.71 \text{ N}$$

$$F_{1y} = 16 \text{ N}$$

$$F_{2x} = 55 \text{ N}$$

$$F_{2y} = 0 \text{ N}$$

$$F_{3x} = 20.5$$

$$F_{3y} = -35.5$$

$$\begin{aligned} \underline{F_x} &= F_{1x} + F_{2x} + F_{3x} \\ &= 27.71 + 55 + 20.5 \\ &= 103.21 \text{ N} \end{aligned}$$

$$\begin{aligned} \underline{F_y} &= F_{1y} + F_{2y} + F_{3y} \\ &= 16 + 0 - 35.5 \\ &= -19.5 \text{ N} \end{aligned}$$

Now,

$$F_x = ma_x$$

$$a_x = \frac{F_x}{m} = \frac{103.21}{120}$$

$$a_x = 0.86 \hat{\text{m s}}^{-2}$$

$$a_y = \frac{F_y}{m} = \frac{-19.5}{120} = -0.162 \hat{j} \text{ N}$$

$$\vec{a} = (0.86 \hat{i} - 0.16 \hat{j}) \text{ m}^2$$

$$|a| = 0.87 \text{ ms}^{-2}$$

$$\theta = -10.5^\circ$$

Q no: 6

$$F_A = 220 \text{ N}$$

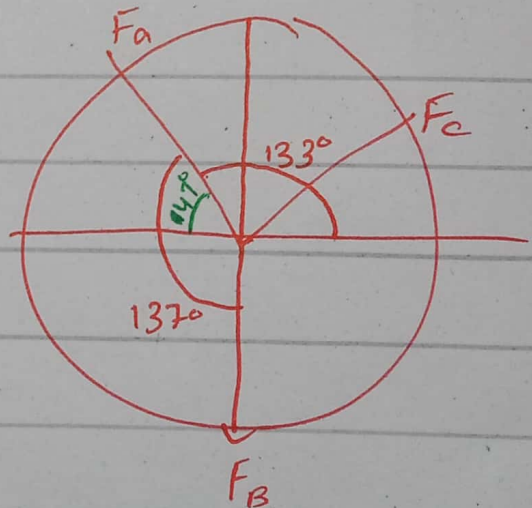
$$F_C = 170 \text{ N}$$

$$\theta_A = 133^\circ$$

$$\theta_C = ?$$

$$\theta_B = 137^\circ$$

$$F_B = ?$$



$$\Sigma F_x = F_A + F_C \text{ (along x-axis)}$$

$$\Sigma F_x = F_A \cos \theta + F_C \cos \theta$$

$$0 = 220 \cos 133^\circ + 170 \cos \theta$$

$$0 = -150 + 170 \cos \theta$$

$$170 \cos \theta = +150$$

$$\cos \theta = \frac{+150}{170}$$

$$\theta_c = 28^\circ$$

Now, For F_B .

$$\begin{aligned} F_B &= F_A \sin \theta + F_c \sin \theta \\ &= 220 \sin 47 + 170 \sin 28 \\ F_B &= 240.7 \text{ N} \end{aligned}$$

Q no: 9

$$m = 0.34 \text{ kg}$$

$$x(t) = -15 + 2t - 4t^3$$

$$y(t) = 25 + 7t - 9t^2$$

$$\text{at } t = 0.7 \text{ sec}$$

$$|F| = ?$$

Solution:

$$\frac{d^2}{dx} x t = 2 - 12t^2$$

$$= -24t = a_x$$

$$\frac{d^2}{dy} y t = 7 - 18t$$

$$= -18 = a_y$$

$$\begin{aligned}
 a_x(t) &= -24t \\
 &= -24(0.7) \\
 &= -16.8
 \end{aligned}$$

$$a_y = -18$$

$$|a| = \sqrt{(-16.8)^2 + (-18)^2}$$

$$|a| = \sqrt{600.6.24}$$

$$|a| = 24.62$$

$$\begin{aligned}
 F &= m a \\
 &= (0.34) (24.62) \\
 &= 8.37 \text{ N}
 \end{aligned}$$

Q no: 13

$$T_w = 98 \text{ N}$$

$$T_1 = 58.8 \text{ N}$$

$$T_2 = 49 \text{ N}$$

$$T_3 = 9.8 \text{ N}$$

$$a_y = g = 9.8$$

$$m_a = ?$$

$$m_b = ?$$

$$m_c = ?$$

$$m_d = ?$$

Solution:
For m_a

$$\Sigma F = T_w - T_1 - F_a$$

$$\Sigma F = T_w - T_1 - m_a g$$

$$0 = T_w - T_1 - m_a g$$

$$m_a g = T_w - T_1$$

$$m_a = \frac{T_w - T_1}{g}$$

$$m_a = \frac{98 - 58.8}{9.8}$$

$$m_a = 4 \text{ kg}$$

For, m_b :

$$m_b = \frac{T_1 - T_2}{g}$$

$$m_b = \frac{58.8 - 49}{9.8}$$

$$m_b = 1 \text{ kg}$$

For m_c :

$$m_c = \frac{T_2 - T_3}{g}$$

$$m_c = \frac{49 - 9.8}{9.8}$$

$$m_c = 4 \text{ kg}$$

For m_d ;

~~$m_d =$~~

$$\Sigma F = T_3 - m_d g$$

$$m_d g = T_3$$

$$m_d = \frac{T_3}{g}$$

$$m_d = \frac{9.8}{9.8}$$

$$m_d = 1 \text{ kg.}$$

Q no: 14

weight = 3 N

upward force = 1 N

a) Magnitude in horizontal

block applying 3 N on floor now 1 N apply on upward so,

Mag of force is Now

$$3 - 1 = 2 \text{ N}$$

b) As 1 N is on upward and 2 N downward so direction is in downward direction.

Q no: 17

mass = 8.5 kg

$$\theta = 30^\circ$$

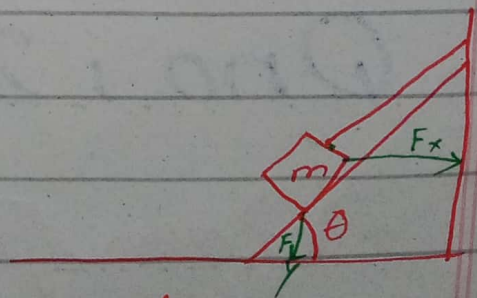
$$a = g = 9.8$$

a) Tension in cord

$$F_{gy} = mg \sin \theta$$

$$= (8.5)(9.8) \sin 30$$

$$= 41.65 \text{ N}$$



b) Normal force acting on block.

$$\begin{aligned} F_{gx} &= mg \cos \theta \\ &= (8.5)(9.8) \cos 30 \\ &= 72.13 \text{ N} \end{aligned}$$

c) when cord cut force on block.

when cord is cutted tension of cord appear as force on block which is 41.67 N

so,

$$\begin{aligned} F &= ma \\ a &= \frac{F}{m} = \frac{41.67}{8.5} \end{aligned}$$

$$a = 4.9 \text{ ms}^{-2}$$

Qno : 21

From graph.

$$a_x = \frac{\Delta v}{\Delta t}$$

$$a_x = \frac{v_{\text{ext}} - v_{\text{xi}}}{t_f - t_i}$$

$$a_x = \frac{8 - 5}{2 - 1} = 3 \text{ m/s}^2$$

$$a_y = \frac{v_{yf} - v_{fi}}{t_f - t_i} = \frac{-8 - (-3)}{2 - 1}$$

$$a_y = \frac{-8 + 3}{1} = -5 \text{ m/s}^2$$

$$\begin{aligned} F_x: F_{xa} &= m a_x \\ &= 2(3) = 6 \text{ N} \end{aligned}$$

$$F_y: F_y = 2(-5) = -10 \text{ N}$$

$$|F| = \sqrt{6^2 + (1-0)^2}$$

$$|F| = \cancel{8.57} \text{ N} \quad 11.66 \text{ N}$$

Direction

$$\theta = \tan^{-1} \left(\frac{-10}{6} \right) = -59.03$$

direction is below the horizontal.
or -59.03° south of east.

Q no : 24

$$m = 2 \text{ kg}$$

$$F_1 = 20 \text{ N}$$

$$F_2 = ?$$

$$a) a = 10$$

$$b) a = 20$$

$$c) a = 0$$

$$d) a = -10$$

$$e) a = -20$$

$$F_2 = ?$$

$$a) a = 10 \text{ ms}^{-2}$$

$$F = (10)(2) = 20 \text{ N}$$

$$\text{if } F_1 = 20$$

$$\text{so } F_2 = 0$$

$$b) a = 20 \text{ ms}^{-2}$$

$$F = (20)(2) = 40 \text{ N}$$

$$\text{if } F_1 = 20$$

$$\text{so, } F_2 = 20$$

$$c) a = 0 \text{ ms}^{-2}$$

$$F = (2)(0) = 0 \text{ N}$$

$$\text{if } F_1 = 20 \text{ N}$$

$$\text{so, } F_2 = -20 \text{ N}$$

$$d) a = -10 \text{ m s}^{-2}$$

$$F = (2)(-10)$$

$$F = -20$$

$$\text{if } F_1 = 20$$

$$\text{so, } F_2 = 0$$

$$e) a = -20 \text{ m s}^{-2}$$

$$F = (2)(-20) = -40$$

$$\text{if } F_1 = 20 \text{ N}$$

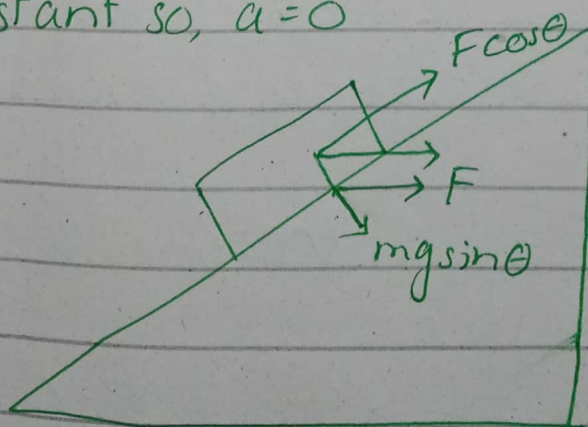
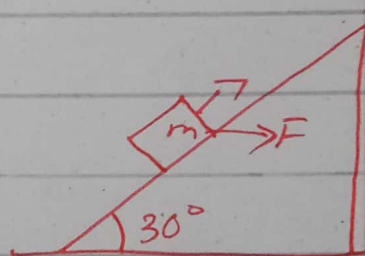
$$\text{so, } F_2 = -60 \text{ N}$$

Q no: 34

$$m = 100 \text{ kg}$$

$$\theta = 30^\circ$$

as speed is constant so, $a = 0$



$$F \cos \theta - mg \sin \theta = ma$$

$$F \cos \theta - mg \sin \theta = m(0)$$

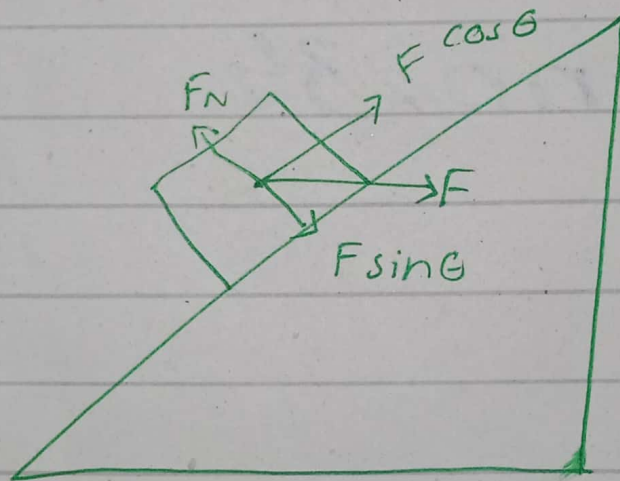
$$F \cos \theta = mg \sin \theta$$

$$F = \frac{mg \sin \theta}{\cos \theta}$$

$$F = mg \tan \theta$$

$$F = (100)(9.8) \tan 30$$

$$F = 565.8 \text{ N}$$



$$F_N - F \sin \theta - \overbrace{mg \cos \theta}^{\text{weight}} = ma$$

$$F_N - (565.8) \sin 30 - (100)(9.8) \cos 30 = 0$$

$$F_N - 289.2 - 848.70 = 0$$

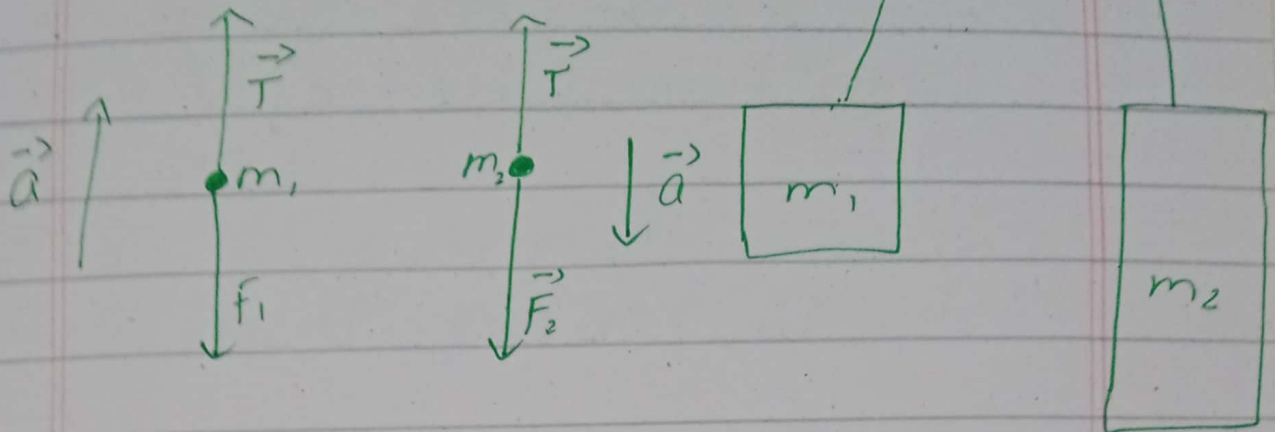
$$F_N = 1137.9 \text{ N} \quad \text{or} \quad 1.13 \times 10^3 \text{ N}$$

$$m_1 = 1.3 \text{ kg} ; m_2 = 2.8 \text{ kg}$$

$$|a| = ?$$

$$T = ?$$

Q no: 51



$$T - m_1 g = m_1 a \quad \text{--- (i)}$$

$$T - m_2 g = -m_2 a$$

$$m_2 g - T = m_2 a \quad \text{--- (ii)}$$

Adding (i) & (ii)

$$T - m_1 g = m_1 a$$

$$T + m_2 g = m_2 g$$

$$(m_2 - m_1)g = (m_1 + m_2)a$$

$$a) \quad a = \frac{(m_2 - m_1)g}{m_1 + m_2}$$

can be start from here

$$a = \left(\frac{2.8 - 1.3}{1.3 + 2.8} \right) g$$

$$a = 3.585 \text{ ms}^{-2}$$

b)

$$T - m_1 g = m_1 a$$

$$T - m_1 g = m_1 \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g$$

$$T = m_1 \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g + m_1 g$$

$$T = m_1 g \left(\frac{m_2 - m_1}{m_1 + m_2} + 1 \right)$$

$$T = m_1 g \left(\frac{m_2 - m_1 + m_1 + m_2}{m_1 + m_2} \right)$$

$$T = \frac{2m_1 m_2 g}{m_1 + m_2}$$

$$T = \frac{2(1.3)(2.8)(9.8)}{1.3 + 2.8}$$

$$T = 17.4 \text{ N}$$

Check-point_1

figure ~~(b)~~, (c), (d), & (e) shows the correct vector addition.

Check-point_2

a) To keep the stationary state $\vec{F}_3$ is of magnitude $2N$ and its direction would be in left.

b) as speed is constant so net force again would be zero, direction of $\vec{F}_3$ is again in left and magnitude would be $2N$ again.

Check-point_3

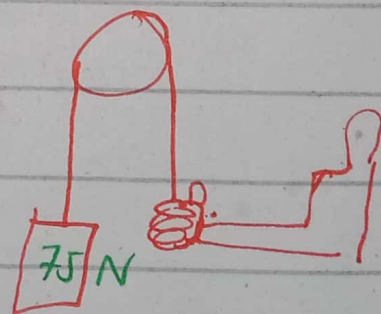
a) at constant speed, the the block is in equilibrium again an the normal force F_n and mg are both equal to each other.

(b) At increasing speed the weight mg in downward remains same but due to increasing acceleration in upward direction an apparent weight is added in it so, to counter this apparent weight magnitude of normal force F_N is greater than mg .

Check_point_4

a) at constant speed.

$$T = 75 \text{ N}$$



b) At increasing speed.

$$T > 75 \text{ N}$$

c) At decreasing speed

$$T < 75 \text{ N}$$